Product Design / Mechanism Design

# Universal Travel Adapter

A compact universal travel adapter that simplifies plug switching through a single-step twist mechanism while optimizing the internal component layout.

Project type

Personal innovation project

Role

Product & mechanism designer, end-to-end

Timeline

2024 - 2025

Team

Individual project

Tools

Fusion 360, SolidWorks, KeyShot, Rapid prototyping


Personal innovation project — patent in progress

02

## The problem

Travelers typically have an adapter which can be specific to regions, a heavy multi-piece adapter set, or a universal block with an array of sliders that confuse you. This problem is a result of losing pieces, being confused about the pins and a heavy housing that covers other outlets.

This project had a mechanical problem of consolidating several plug standards into one unit.


Existing universal adapters trade compactness for flexibility, often doing neither well.

03

## Research & insight

> Users think in destinations, not pin standards — the interaction must hide the technical complexity of plug types.
>
> Slider-based universal adapters fail because each pin set moves independently; users want one confident action.
>
> Volume is the enemy: every cubic centimetre saved matters in carry-on travel, and oversized housings block neighbouring sockets.

> Safety interlocks can double as tactile feedback, turning a constraint into a usability feature.

04

## Design direction

Design Direction The luminaires were designed with two optical systems. These include the direct module used to project light into the road and the indirect module which illuminates pedestrian walkways using low glare light.

### Dual optical architecture

Direct lighting for the roadway, indirect for pedestrian zones, engineered as separate systems inside a single housing.

### Independent zone control

Each zone supports its own intensity and color-temperature profile, enabling time-of-day and occupancy-based tuning.

### Modular serviceability

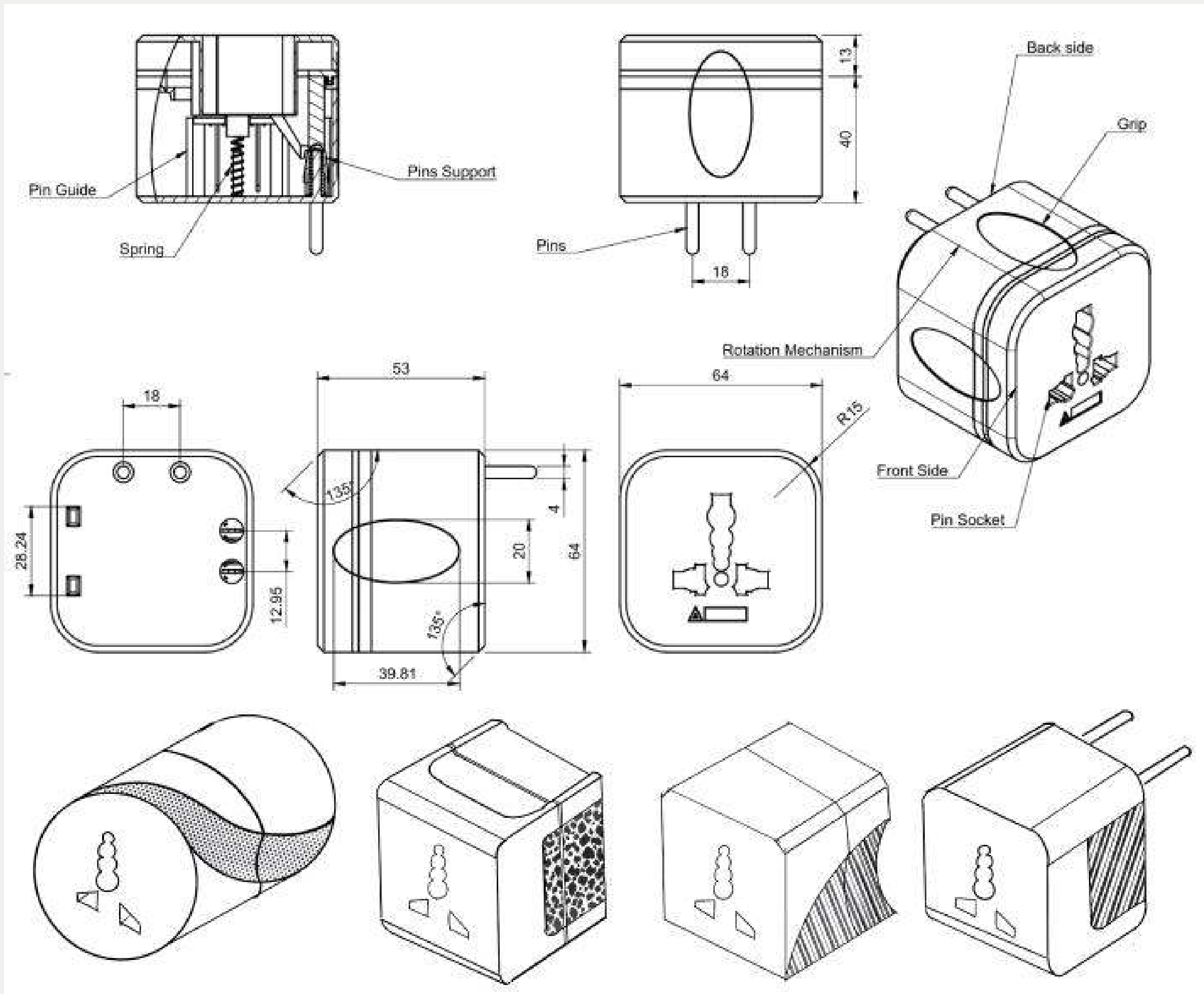
The housing decomposes into replaceable modules so a failed zone never requires replacing the whole fixture.

05

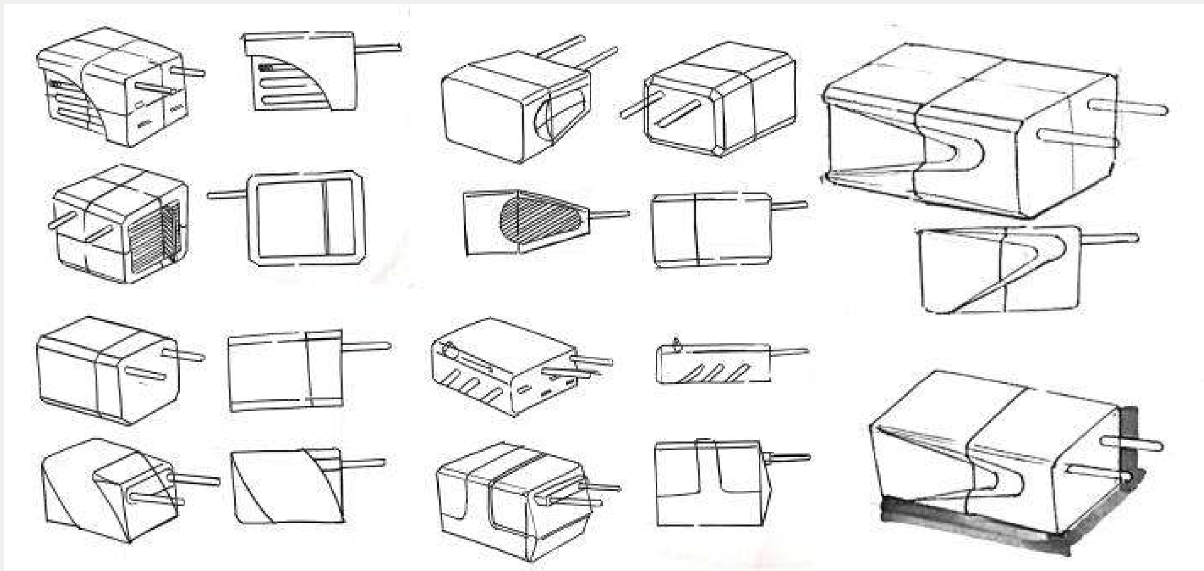
## Ideation & exploration

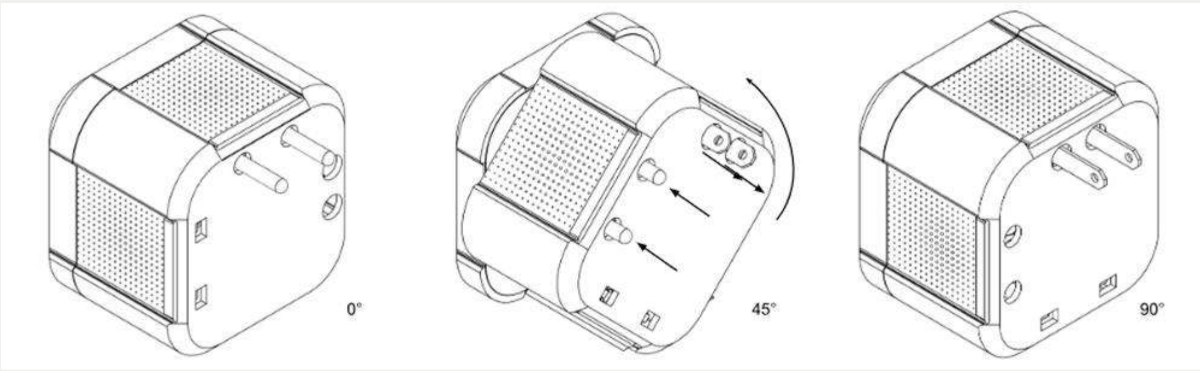
Exploration moved from slider arrays and flip-out pin clusters toward rotary architectures. Early slider concepts were rejected because they preserved the exact interaction problem they were meant to solve; flip-out clusters were rejected for part count and durability risk.

- Configuration studies mapping US, UK, EU, and AU pin geometries onto shared axes
- Form studies balancing grip diameter against internal volume
- Interaction studies on twist direction, detent strength, and one-handed operation



Pin Guide
Spring
Pins Support
Pins
18
13
40
Back side
Grip
Rotation Mechanism
Front Side
Pin Socket
18
28.24
12.95
53
135°
4
20
64
135°
39.81
64
R15



Configuration and form studies converging on a rotary architecture.

06

# Mechanisms & system diagrams

The mechanism chapter of this project is where the design was won or lost. Component breakdowns and exploded views were used to verify that every configuration could share the same electrical contacts and that the rotary selector could physically index each pin set.

01

## Exploded component layout

Verified stacked arrangement of contacts, selector drum, and housing halves to minimise overall volume.

02

## Rotary indexing sequence

Mapped detent positions to plug configurations so each click state corresponds to exactly one regional standard.

03

## User interaction sequence

Twist, confirm the detent click, plug in — a three-step sequence replacing multi-slider setup.

07

# Prototyping & iteration

Rapid prototypes were used to validate the twist action, detent feel, and overall volume. Each printed iteration informed tolerance adjustments in the selector drum and housing wall thicknesses.

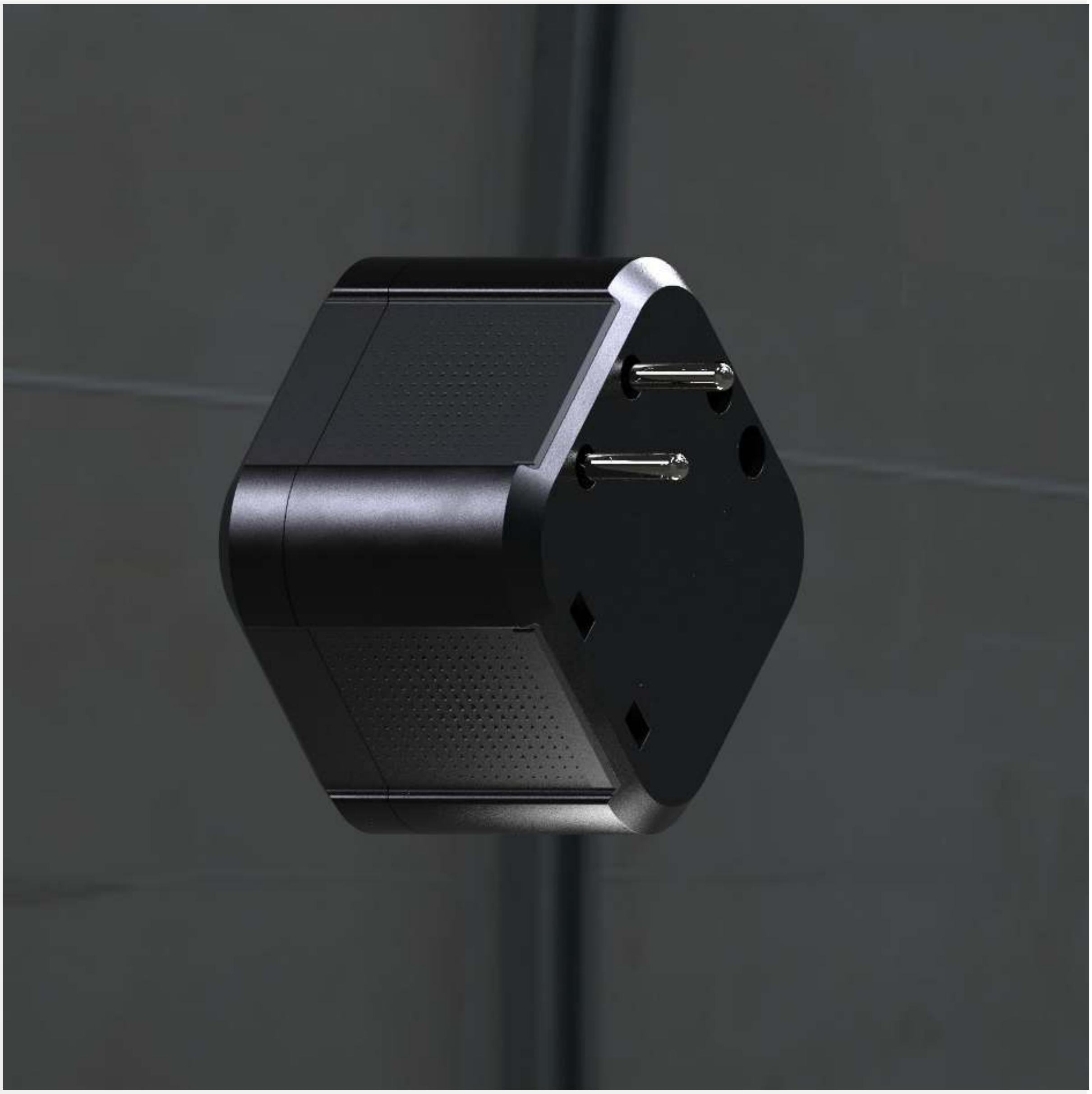
- Fit and volume checks against carry-on travel constraints
- Detent strength tuning for confident, one-handed switching
- Wall-thickness and rib optimisation to keep the body rigid at reduced size

08

# Final solution

The final concept is a single enclosed adapter body with a rotary selector: one twist moves between plug configurations, with each position confirmed by a tactile detent. Internal components share contact paths, keeping the housing compact enough to leave neighbouring sockets free.

Single-step twist mechanism for all supported plug configurationsTool-less, enclosed architecture with no detachable partsOptimised internal component layout for minimum volumeTactile detent feedback doubling as a safety interlock



Final concept — one compact body, one twist, every region.

09

## Outcome & metrics

> Patent in progress for the unified plug-switching mechanism
>
> Improved compactness through internal component optimization
>
> Simplified user interaction from multi-slider setup to a single twist

10

## Reflection

## What I learned

Mechanism design is interaction design. The tactile quality of the twist — detent strength, rotation angle, end stops — shaped the user experience as much as the architecture itself.

## What I'd improve next

Next steps are electrical safety certification testing and design-for-manufacture refinement of the contact stamping and drum tolerances.

## What this says about my approach

This project shows how I work from the detail outward: a single mechanism decision, rigorously resolved, can define an entire product.

Urban Infrastructure / Lighting Design

# Dual-Zone Smart LED Streetlight

A dual-zone streetlight that separates roadway and pedestrian illumination into independently controlled direct and indirect lighting systems within one housing.

Project type

Graduation project

Role

Product / Industrial Design

Timeline

2025 - 2026

Team

Lighting design team

Tools

Fusion 360, KeyShot, Adobe Illustrator, Prototyping

Graduation project

02

## The problem

The urban streets have to cater to two very different groups simultaneously: drivers requiring high-contrast lighting on the roads and pedestrians needing comfortable eye-level lighting without glare. Conventional streetlights can accommodate both from one single source and one optic only.

In essence, this results in a compromise in everything from glare to pedestrians, to wastage in terms of spill light and energy usage for zones that might not have any people in them at all.

Single-source streetlights force one optic to serve two incompatible lighting tasks.

03

## Research & insight

> Road and pedestrian lighting require different optical needs — one luminaire must address both zones independently.
> Adaptive lighting can reduce unnecessary output by responding to ambient light and pedestrian movement.
> Maintenance creates a significant opportunity — tool-free access can reduce service effort and downtime.
> Modularity enables independent replacement of drivers, LED modules, sensors, and communication components as they reach different lifespans.

04

## Design direction

The design direction centered on integrating two lighting zones into one modular luminaire while prioritizing independent control, glare reduction, thermal management, and simple maintenance.

**Dual-zone optical architecture**

Separate road and pedestrian optics provide targeted illumination for each zone.

**Independent adaptive control**

PIR and LDR inputs allow each lighting zone to respond to changing conditions.

**Tool-less serviceability**

Make critical components accessible without specialist tools.

**Shared thermal spine**

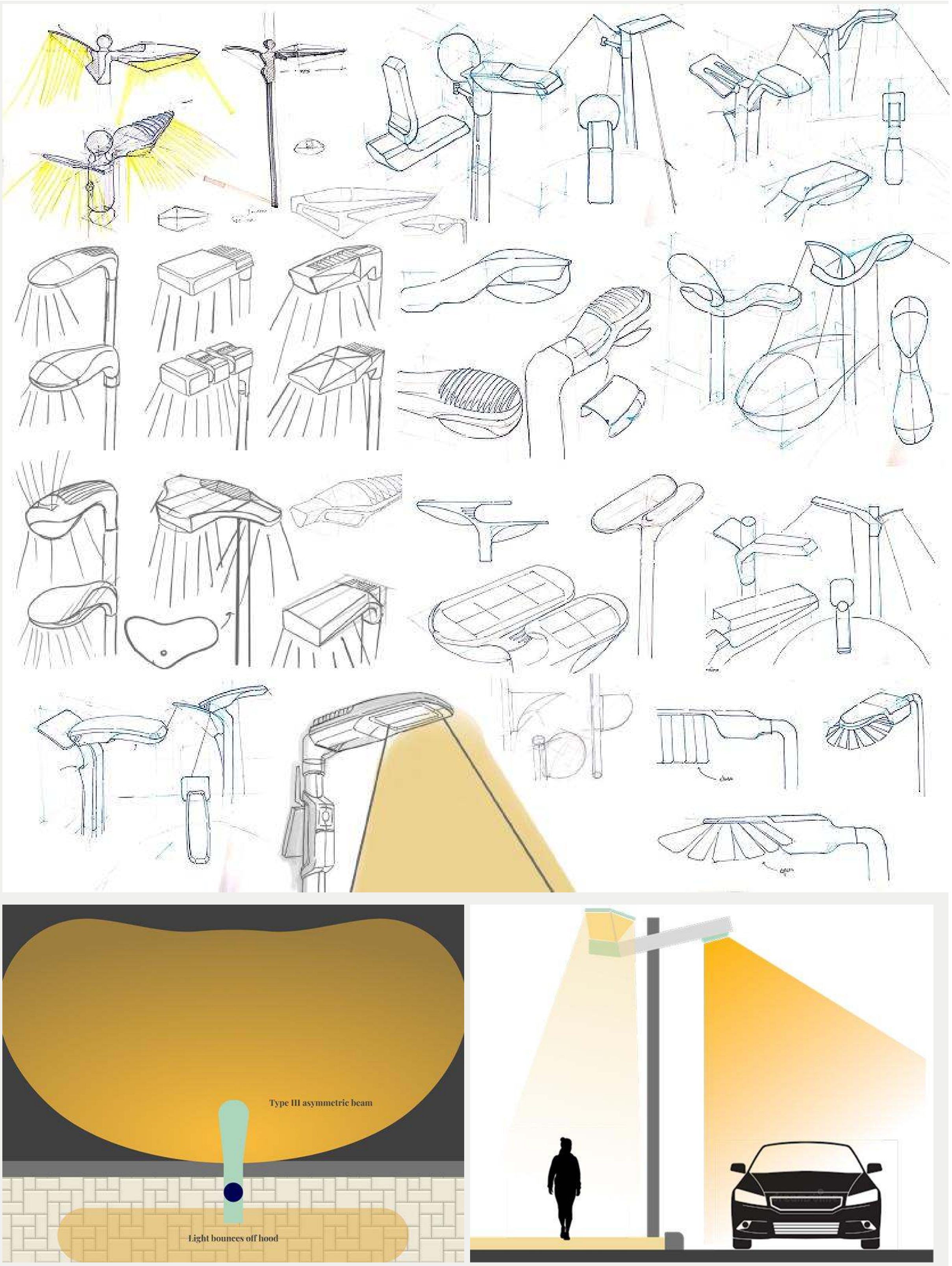
Use one aluminium structure to manage heat across both zones.

05

## Ideation & exploration

Exploration moved from multiple housing and optical configurations toward a unified dual-zone architecture. Concepts were compared for optical separation, maintenance access, structural integration, and overall form before selecting the most feasible direction.

- Dual-zone housing studies balancing road and pedestrian lighting
- Optical studies exploring direct and indirect light distribution
- Form studies integrating two lighting zones into one housing
- Serviceability studies focused on tool-free component access



Housing and optic layout exploration during ideation.

06

## Mechanisms & system diagrams

The system architecture integrates dual-zone optics, independent LED control, sensing, thermal management, and IoT communication. Component breakdowns define how each subsystem works together while maintaining modular access for service and replacement.

01

### Dual-zone optical architecture

Separate road and pedestrian optical chambers provide dedicated illumination from one housing.

02

### Adaptive sensing & control

PIR and LDR inputs independently control brightness through the dual-channel control board.

03

### Modular service architecture

Independent LED, driver, control, IoT, and sensor modules enable tool-free replacement.

07

## Prototyping & iteration

Functional prototyping focused on validating the dual-zone lighting architecture, sensor placement, modular access, and overall housing configuration. Iterations helped refine the component layout, maintenance approach, and integration of the optical and thermal systems.

- Validation of road and pedestrian optical separation
- Testing of tool-free gear-tray access
- Refinement of modular component arrangement
- Validation of sensor placement and adaptive lighting behavior

08

## Final solution

The final concept is a single integrated dual-zone luminaire that separates road and pedestrian lighting within one housing, using independent optics, adaptive sensing, and modular tool-free access for maintenance.

Dual-zone optical system for road and pedestrian illumination
Tool-free modular architecture for quick component replacement
Independent PIR + LDR adaptive lighting control
Shared aluminium heat-sink spine for both lighting zones

One housing, two independent lighting systems.

09

## Outcome & metrics

> Developed a unified dual-zone lighting system within a single luminaire housing
>
> Simplified maintenance through modular, tool-free component access
>
> Enabled independent adaptive control for road and pedestrian lighting
>
> Integrated sensing and IoT connectivity for responsive lighting management

> Established a modular architecture for easier component replacement and upgrades

10

# Reflection

## What I learned

User research revealed that pedestrian glare and comfort can matter as much as illumination levels. I also learned that modularity must be considered from the beginning because serviceability directly affects product architecture.

## What I'd improve next

I would conduct more rigorous photometric and environmental testing, particularly to quantify energy performance, optical distribution, thermal behavior, and long-term outdoor durability.

## What this says about my approach

I approach product design as a **system problem**, connecting user needs, mechanisms, materials, serviceability, and technology before resolving the final form.

Urban Infrastructure / Lighting Design

# Tool-less Modular Outdoor Luminaire System

A modular outdoor luminaire architecture designed around tool-less maintenance, replaceable lighting modules, and long-term serviceability.

Project type

Industrial Design · Outdoor Lighting · Modular Product Design

Role

Industrial Design Intern

Timeline

6-month industry internship · 2025

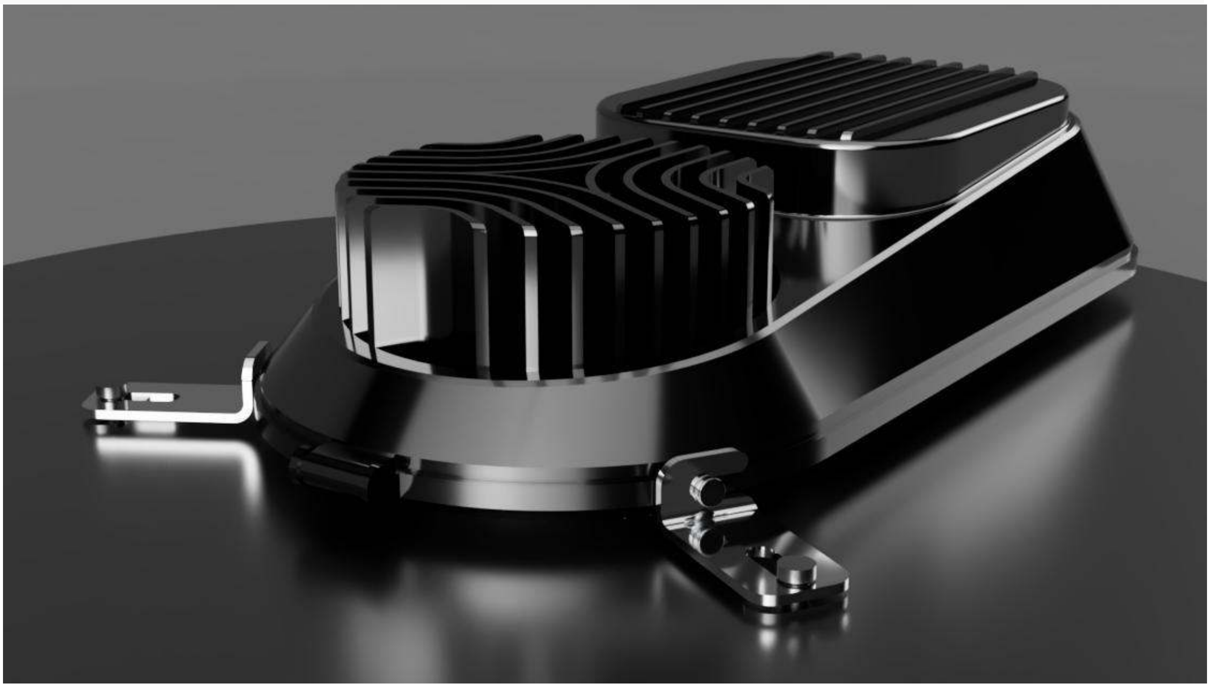
Team

Industry Mentors: Sachin Kumar Garg & Akash Singh

University Mentor: Dr. Arunachalam

Tools

Fusion 360 · CAD Modelling · KeyShot · Sketching · Rapid Prototyping



Quick-maintenance modular lighting module developed for outdoor luminaire applications.

## The problem

Outdoor luminaire maintenance often requires trained technicians and tools, increasing service time, cost, and the risk of damage during repair. The design challenge was to create a modular lighting system that could be opened, removed, and serviced without tools while remaining mechanically reliable and suitable for outdoor conditions.

Servicing outdoor luminaires requires tools, time, and difficult access.

03

## Research & insight

> **Maintenance should be accessible.**
>
> Basic module replacement should not depend entirely on specialised tools or trained technicians.
>
> **Modularity can extend product life.**
>
> Replaceable LED and driver modules can support repair, upgrading, and long-term use.

> **Tool-less mechanisms need intuitive interaction.**
> Latches, sliders, pins, and twist mechanisms offered references for simpler maintenance operations.

04

## Design direction

The design direction focused on creating a modular luminaire architecture where the lighting module could be accessed through simple mechanical actions while maintaining compatibility with the larger fixture system.

### Tool-less Access

Develop a module that can be opened, removed, and serviced without conventional tools.

### Modular Architecture

Separate the lighting components into replaceable modules to support repair and future upgrades.

### Mechanical Simplicity

Explore latches, pins, thumb screws, and twist interfaces that reduce maintenance complexity.

### System Compatibility

Maintain consistent geometry, proportions, and interfaces so the module integrates with the luminaire family.

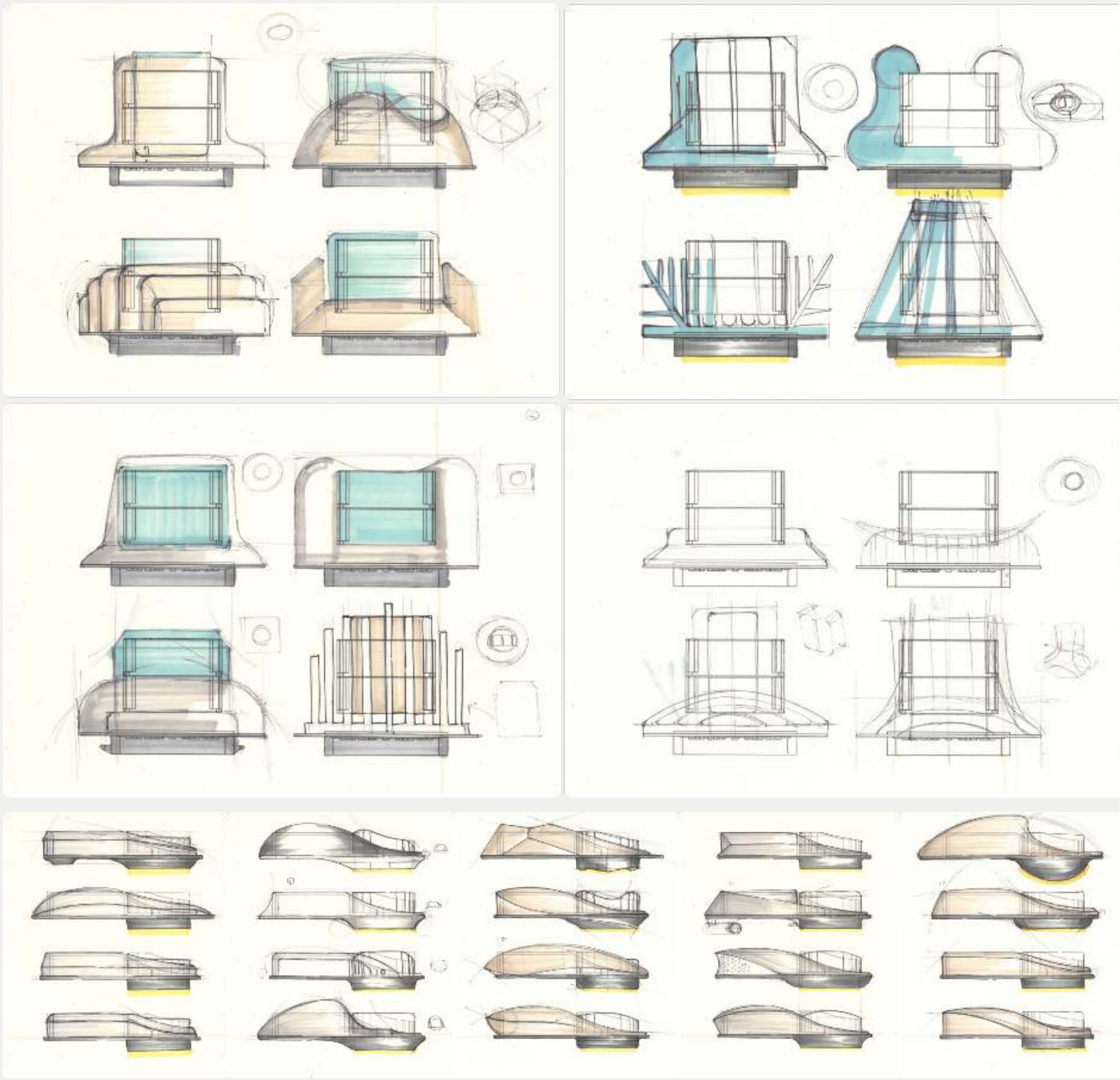
05

# Ideation & exploration

Multiple mechanical and configuration directions were explored for accessing the lighting module. Concepts included quick-release latches, clips, twist attachments, thumb-screw interfaces, takedown pins, and hinged arrangements. Different top- and bottom-placement configurations were also considered before refining the module forms.

The exploration connected mechanism studies with physical appearance development. Three selected module types were further developed, with attention to geometry, proportions, handling, and integration into the luminaire system.

- Explored quick-release latch and clip-based module attachment concepts.
- Studied twist, thumb-screw, takedown-pin, and hinged mechanisms.
- Developed alternative top- and bottom-mounted module configurations.
- Refined selected forms for visual and system integration.



Housing exploration during ideation.

06

## Mechanisms & system diagrams

The luminaire was developed as a layered system containing housing, gasket, driver, heat sink, PCB, reflector, glass, and associated mechanical interfaces. Different access mechanisms translated simple manual actions into module removal.

01

### Latch mechanism

The user turns the latch to unlock the module before pulling the lighting module upward.

02

### Twist mechanism

A thumb screw is loosened before the module is rotated approximately 55° and lifted.

03

### Module Architecture

Top and bottom housings integrate the gasket, driver, heat sink, PCB, reflector, glass, and fastening elements.

04

### Replaceable Light Engine

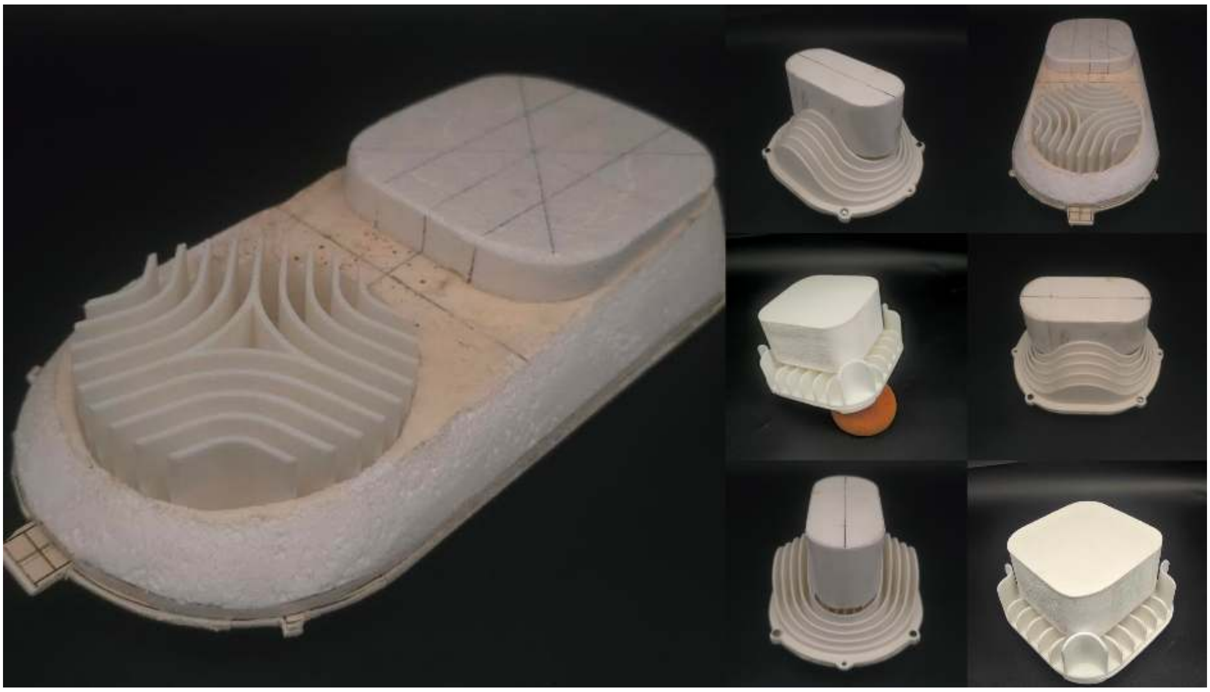
The LED module and driver are treated as serviceable elements within the broader luminaire architecture.

07

## Prototyping & iteration

Appearance prototypes were developed to evaluate form, proportions, physical presence, and visual integration before functional mechanism development. A 1:1 prototype was then created to assess actual scale, handling, fit with the luminaire body, and maintenance ergonomics. Refinement focused on geometry, boundaries, surface transitions, and system compatibility.

- Created five appearance prototypes to explore physical form and proportions.
- Built a 1:1 prototype to evaluate scale and handling.
- Refined geometry and surface transitions for system integration.
- Conducted visual and dimensional evaluation of the final module forms.



08

## Final solution

The final direction developed a modular outdoor luminaire system with replaceable lighting modules and multiple tool-less access concepts. The system combines a structured internal architecture with simplified mechanical interfaces, creating a foundation for repairable and upgradeable outdoor lighting products.

- Tool-less module access and removal
- Replaceable LED and driver architecture
- Integrated housing, sealing, and thermal components
- Multiple modular interface configurations

Tool-less Modular Outdoor Luminaire Housing.

09

## Outcome & metrics

> Three modular mechanism directions were developed and refined.
>
> Five appearance prototypes explored the module's form and visual identity.
>
> A 1:1 prototype validated physical scale and handling considerations.
>
> The final prototype established a consolidated form for further functional development.

10

## Reflection

### What I learned

I learned how industrial lighting design depends on the relationship between form, mechanisms, thermal components, sealing, optics, and service requirements.

Designing the exterior became an exercise in coordinating the complete product architecture rather than treating form as an isolated layer.

## What I'd improve next

I would further develop and functionally validate the selected tool-less mechanisms under realistic outdoor-use conditions. More detailed testing of sealing, mechanical durability, thermal performance, and repeated maintenance cycles would strengthen the system.

## What this says about my approach

The project reflects my approach of connecting industrial design with mechanical and technical thinking. I use form exploration alongside system architecture and mechanism studies to create products that are not only visually coherent but also practical to maintain and develop further.

Mobility Systems / Human Factors

# Hybrid-Electric Medical Vehicle

An all-terrain electric medical stretcher designed for stable patient evacuation across rough and uneven terrain.

Project type

Medical Product Design

Role

Mobility & human-factors designer

Timeline

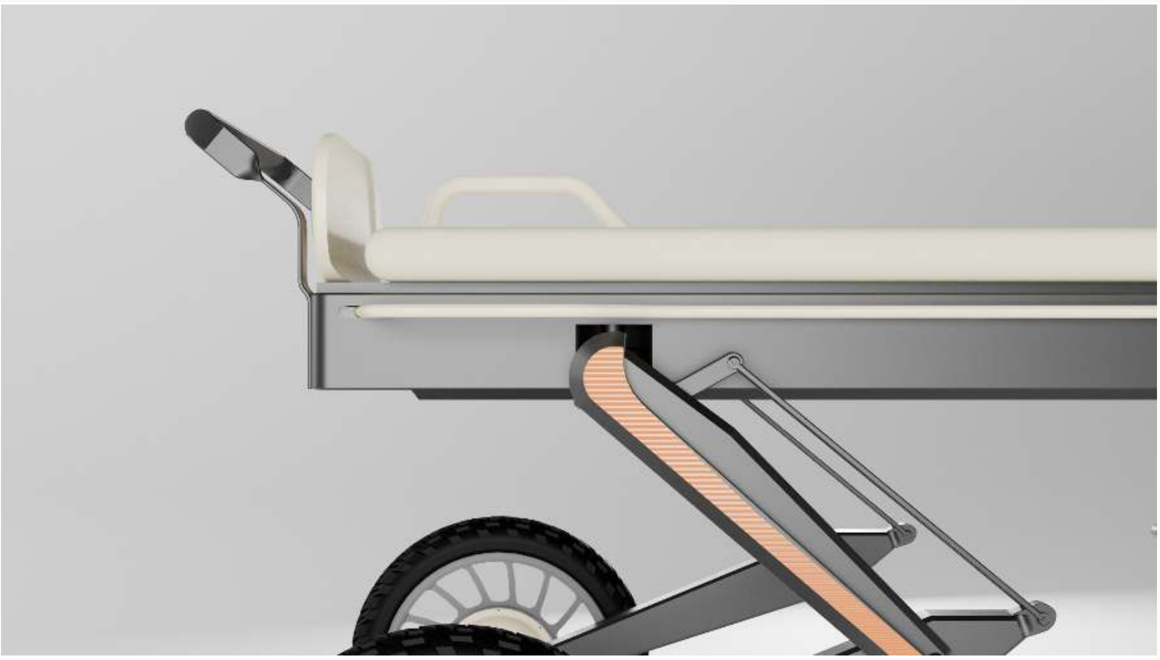
2024 - 2025

Team

Team Project

Tools

Rhino, Fusion 360, SolidWorks, KeyShot,



Emergency medical evacuation concept

02

## The problem

Traditional stretchers struggle in rural and remote rescue environments where rocky, muddy, uneven, or steep terrain creates instability, operator fatigue, poor traction, and patient discomfort. The design challenge was to create a mobile medical stretcher that could maintain patient stability while improving terrain adaptability and reducing physical effort during evacuation.

When stretchers are not made for off-road evacuation, patients and operators suffer.

03

## Research & insight

> Stability is critical - Stretcher instability was identified as a major field problem during movement across uneven terrain.
> Terrain demands adaptability - Rocky, muddy, snowy, sandy, and forest paths require high clearance and dedicated suspension.
> Suspension affects patients - Tracked and rigid approaches can transmit terrain vibration directly to the patient, making suspension quality critical.
> Operator effort matters - Electric assistance and intuitive controls can reduce the physical effort required during emergency transport.

04

## Design direction

The design direction combined large all-terrain wheels, independent suspension, high ground clearance, and a robotic quadruped-inspired linkage to stabilize the stretcher while reducing operator effort.

### Adaptive Suspension

Use independent mechanical linkages to absorb terrain variation and maintain platform stability.

### High Clearance

Create sufficient clearance and optimized wheel geometry for obstacles, slopes, and uneven paths.

### Assisted Mobility

Use electric drive and intuitive controls so one attendant can operate the stretcher.

### Patient Stability

Prioritize a stable bed platform and shock absorption to minimize unwanted patient movement.

05

Multiple configurations were considered around the core requirements of terrain clearance, stability, ergonomics, suspension quality, and structural durability. The evaluated directions included a conventional four-wheel stretcher, quadruped-inspired suspension, rigid high-clearance frame, tracked crawler, and pneumatic scissor-lift base.

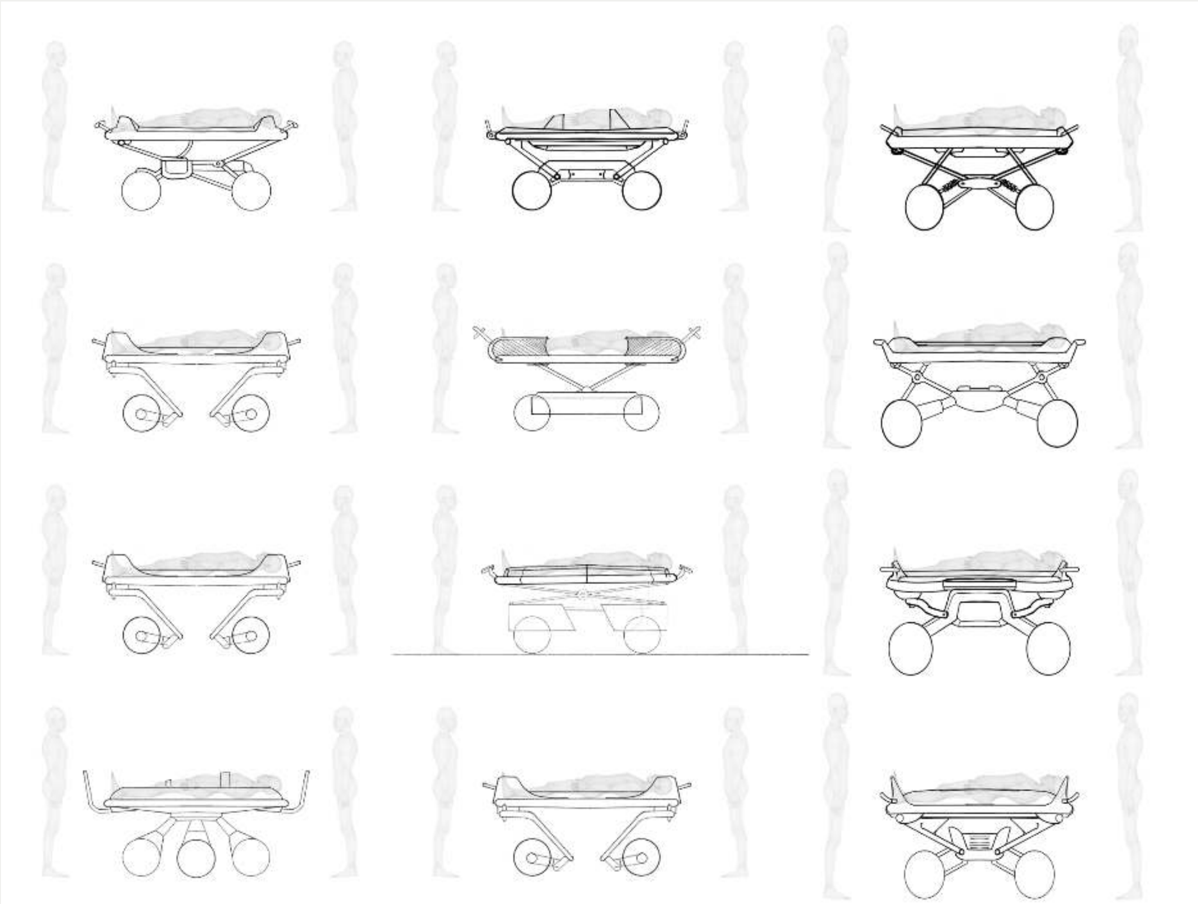
The quadruped-inspired direction was selected because its separate wheel linkages addressed both uneven-terrain stability and suspension requirements rather than focusing only on traction or clearance.

**Conventional four-wheel layout** — established baseline for comparison.

**Rigid high-clearance frame** — prioritized durability and clearance.

**Tracked crawler base** — explored extreme terrain mobility.

**Quadruped-inspired suspension** — combined clearance, stability, and suspension.



Vehicle and stretcher studies exploring isolation strategies.

06

## Mechanisms & system diagrams

MonoHaul uses a single electric drive system, large all-terrain wheels, and a linkage-based suspension architecture. The suspension connects the wheels to the main platform, allowing the stretcher to adapt to uneven terrain while keeping the patient platform more stable.

01

### Quadruped-Inspired Linkage

Separate wheel linkages allow the platform to adapt to uneven ground while maintaining stability.

02

### Spring Suspension

Hard and soft springs provide progressive terrain absorption while supporting different impact conditions.

03

### Electric Drive

A 48V BLDC motor provides powered movement, supporting operation on inclines up to 30°.

04

### Assisted Control

A 48V BLDC motor provides powered movement, supporting operation on inclines up to 30°.

07

## Prototyping & iteration

The supplied material documents concept evaluation and system development, but does not provide specific physical prototype counts, test measurements, or iteration results. The documented evaluation focused on comparing suspension and mobility architectures against the core design criteria.

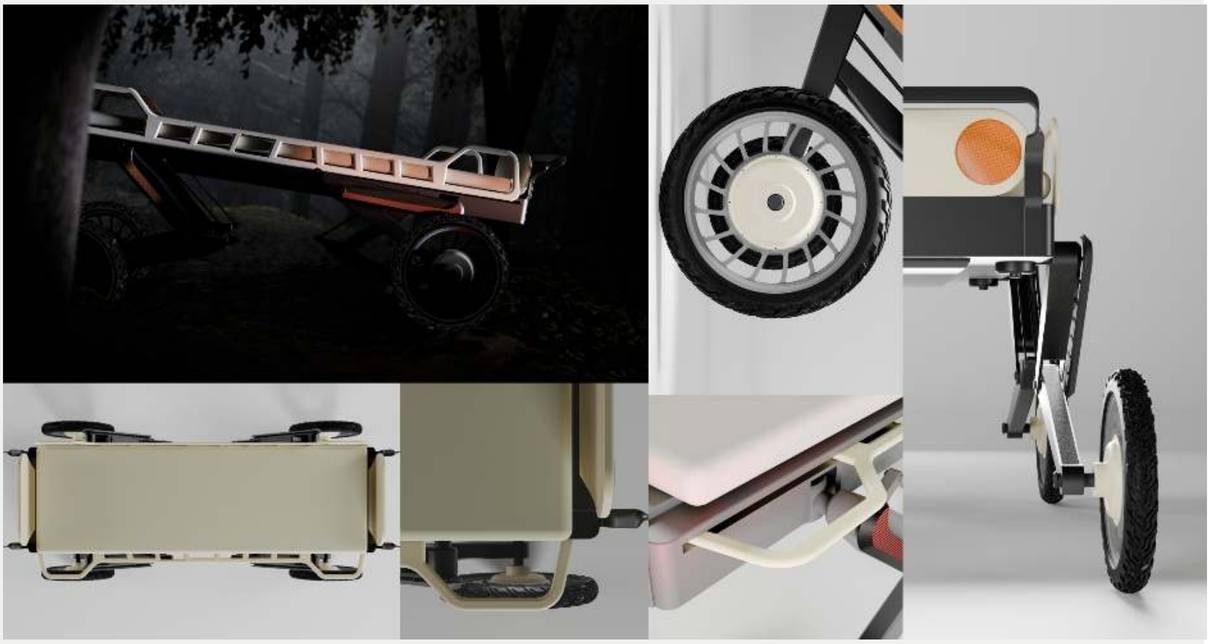
- Compared alternative suspension and mobility architectures.
- Evaluated terrain clearance, stability, ergonomics, suspension, and durability.
- Selected the quadruped-inspired architecture from a 25-point comparison.
- Developed the final concept around all-terrain mobility and patient stability.

08

# Final solution

MonoHaul is an electrically driven, all-terrain medical stretcher designed for emergency transport across rough environments. Its robotic-inspired suspension, large wheels, high ground clearance, and assisted controls combine terrain adaptability with improved patient stability and reduced operator effort.

Quadruped-inspired suspension architecture
Large all-terrain wheels with high clearance
Single-attendant electric mobility
Stable platform for rough-terrain evacuation



stretcher engineered as one stability system.

09

## Outcome & metrics

- Reduced patient and operator strain through adaptive suspension and stretcher design
- Integrated vehicle-stretcher system architecture replacing piecemeal evacuation setups

# Reflection

## What I learned

The project showed how mobility, suspension, patient stability, and operator ergonomics must be treated as one connected system. Solving terrain mobility alone was insufficient; the suspension architecture had to protect both the patient and the operator.

## What I'd improve next

I would validate the suspension and linkage architecture through physical terrain testing, particularly under different loads and slopes. Further testing could also evaluate actual patient comfort, maneuverability, and control behavior.

## What this says about my approach

The project reflects a systems-oriented product design approach where mechanical architecture drives the form. Rather than optimizing one feature, I explored how suspension, mobility, control, safety, and ergonomics could work together as one product.

Medical Design / Human Factors

# Osteoarthritis Mobility Aid

OsteoMove is a lightweight mobility aid designed to support safer walking, exercise, stability, and independence for people with osteoarthritis.

Project type

Special-needs product design

Role

Product designer

Timeline

2023-2024

Team

Individual project

Tools

Ergonomic analysis, Sketching, CAD modelling, Prototyping, Rendering, Welding

Special-needs product design

## The problem

People with osteoarthritis experience joint pain, stiffness, reduced motion, weakness, and instability that can make walking and prescribed exercise difficult. The design challenge was to create a compact, controllable mobility aid that supports movement while reducing strain and improving stability.

Standard aids route force through the very joints their users need to protect.

03

## Research & insight

> **Joint stiffness limits movement.** Osteoarthritis can cause pain, stiffness, reduced range of motion, weakness, and joint instability.
>
> **Exercise remains important.** A rheumatologist recommended activities including walking, cycling, resistance exercises, Tai Chi, and Yoga.
>
> **Walking support must prevent falls.** Field research identified fall prevention as an important additional feature for the mobility aid.
>
> **Support should preserve independence.** The product brief emphasized easier movement, regular activity, safety, comfort, and greater independence.

04

## Design direction

The design direction combined walking assistance with exercise-oriented movement, prioritizing stability, ergonomic support, controlled resistance, compactness, and safer interaction.

### Stability First

A self-balancing structure and anti-tip feature support safer movement and reduce instability concerns.

### Controlled Movement

Adjustable wheel resistance allows the mobility aid to support exercise alongside everyday walking.

### Ergonomic Support

Supported grips and seating elements provide a more comfortable and controlled user position.

### Compact Construction

A lightweight, compact structure with a collapsible handle supports easier handling and storage.

05

## Ideation & exploration

The exploration began with multiple side-view concepts before progressively developing the selected direction with greater structural and functional detail. The final configuration brought together the main frame, seat, back support, wheels, and collapsible handle.

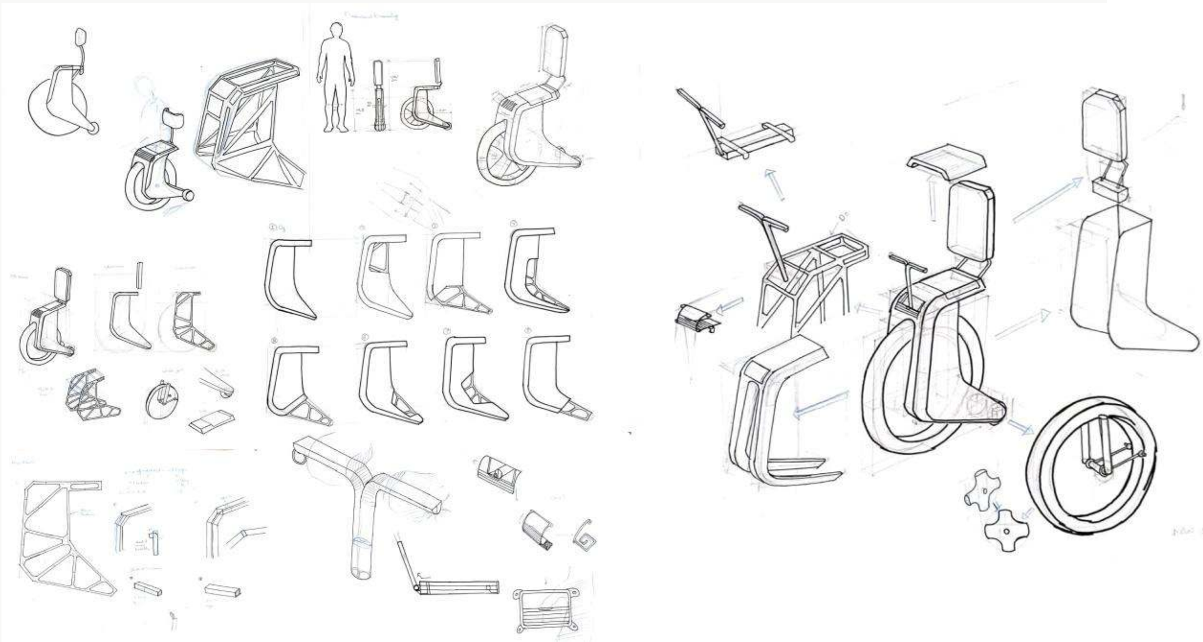
The selected concept was then developed through CAD modelling, rendering, dimensional studies, and physical prototyping. The exploration focused on integrating support and mobility into a single product rather than treating the walking aid as only a support frame.

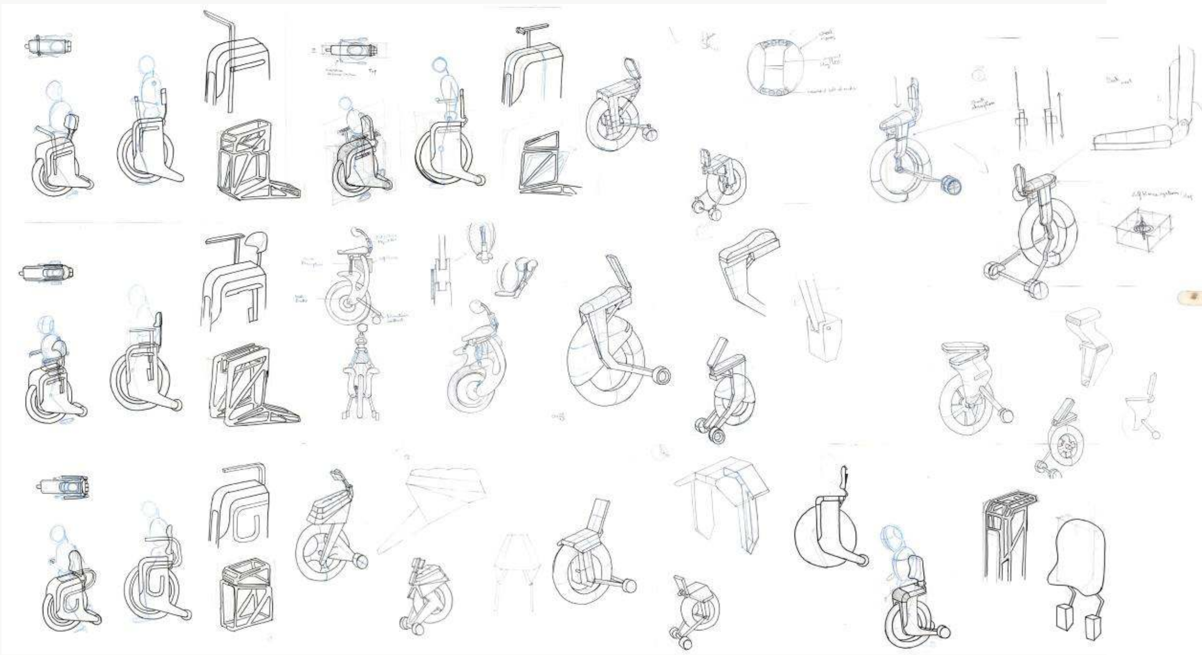
Explored multiple side-view configurations for the overall product form.

Developed detailed studies of the selected concept.

Integrated seating and back-rest support into the structure.

Studied wheel and frame dimensions before prototyping.





Support and grip geometry studies.

06

# Mechanisms & system diagrams

Ergonomic studies defined the contact points, angles, and adjustment ranges that accommodate varying joint mobility across users.

01

## Main Frame

The main frame creates the structural foundation connecting the seat, support elements, and wheel assembly.

02

### Wheel Support

A main wheel and two supporting wheels create the product's mobility and stability structure.

03

### Back Support

The back-rest support and suspension provide additional physical support while using the aid.

04

### Collapsible Handle

The collapsible handle contributes to the product's compact configuration and handling.

07

## Prototyping & iteration

The prototype was developed through metal-rod cutting, joining, frame construction, painting, assembly, and trials. The process translated the selected concept into a physical structure and allowed the overall configuration to be tried out.

- Constructed the basic frame structure from metal rods.
- Joined structural elements according to the selected configuration.
- Integrated wheels and assembled the product structure.
- Tried the assembled prototype to evaluate the overall configuration.

08

# Final solution

osteoMove is a mobility aid designed for people with osteoarthritis who need walking support while remaining active. Its structure combines ergonomic support, seating, multiple wheels, adjustable features, and stability-oriented elements into a compact product.

Cardiovascular exercise support

Free arms and legs during movement

Adjustable resistance and height

Self-balance and anti-tip stability

Support that works with compromised joints, not against them.

09

## Outcome & metrics

> Integrated walking assistance with exercise-oriented functionality.
>
> Combined seating, back support, and mobility into one system.
>
> Developed a physical frame prototype from the selected concept.
>
> Incorporated stability, shock absorption, and anti-tip requirements.

> Created a compact mobility-aid configuration for independent movement.

10

## Reflection

### What I learned

This project showed me that designing for mobility requires balancing physical support with independence. Understanding joint limitations and exercise needs helped me think beyond a conventional walking aid and consider movement, stability, comfort, and activity together.

### What I'd improve next

I would further validate the product with users and clinicians, especially the adjustable resistance, balance, shock absorption, seating, and anti-tip features. More physical testing could also help refine the structure and overall ergonomics.

### What this says about my approach

The project reflects an approach that connects research, user needs, mechanical thinking, and physical prototyping. I tend to explore form through function, using the product's structural and interaction requirements to guide the final design direction.

Design Research / Systems Mapping

# Systems Thinking — ZP Schools

A systems map of Zilla Parishad school operations that surfaced hidden friction between stakeholders and pointed to leverage points for intervention.

Project type
Field research project

Role
Design researcher

Timeline
2023

Team
Research team project

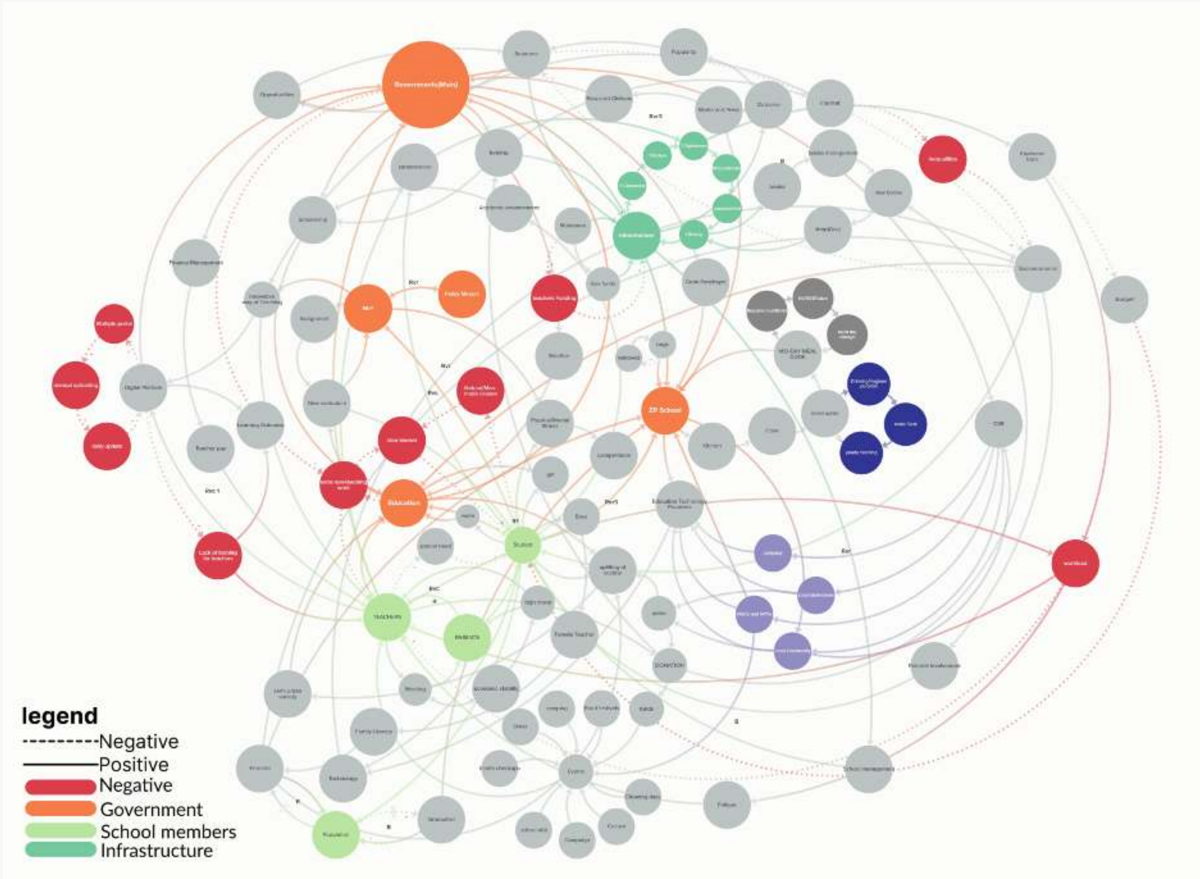
Tools
Systems mapping, Field research, Stakeholder interviews, Causal Loop Diagrams, Iceberg Model, Behaviour Over Time Graphs, Kumu

Field research project

02

## The problem

ZP schools serve rural and low-income communities but operate within interconnected challenges involving limited resources, infrastructure, teacher workload, attendance, technology access, socioeconomic conditions, and community participation. The design challenge was to understand where interventions could create meaningful change within the wider education system, rather than treating individual problems in isolation.




Field research across schools revealed friction no single stakeholder could see alone.

03

## Research & insight

> Teacher commitment compensates for resource limitations, but increases individual workload.

> Digital learning creates opportunities, while device access and technical issues limit its reliability.
> Weather and natural conditions can disrupt school hours and course completion.
> Student attendance and motivation are influenced by wider socioeconomic and family conditions.

04

## Design direction

The project shifted from solving isolated school problems toward identifying relationships, feedback loops, and leverage points where relatively small interventions could influence the broader system.

### Map the Whole System

Understand relationships between academic, administrative, support, regulatory, and community systems before defining interventions.

### Find Leverage Points

Focus interventions on system conditions such as technology reliability and limited school funds rather than only addressing visible symptoms.

### Build on Existing Strengths

Use teacher dedication, student participation, community involvement, and existing school activities as foundations for change.

### Create Positive Feedback

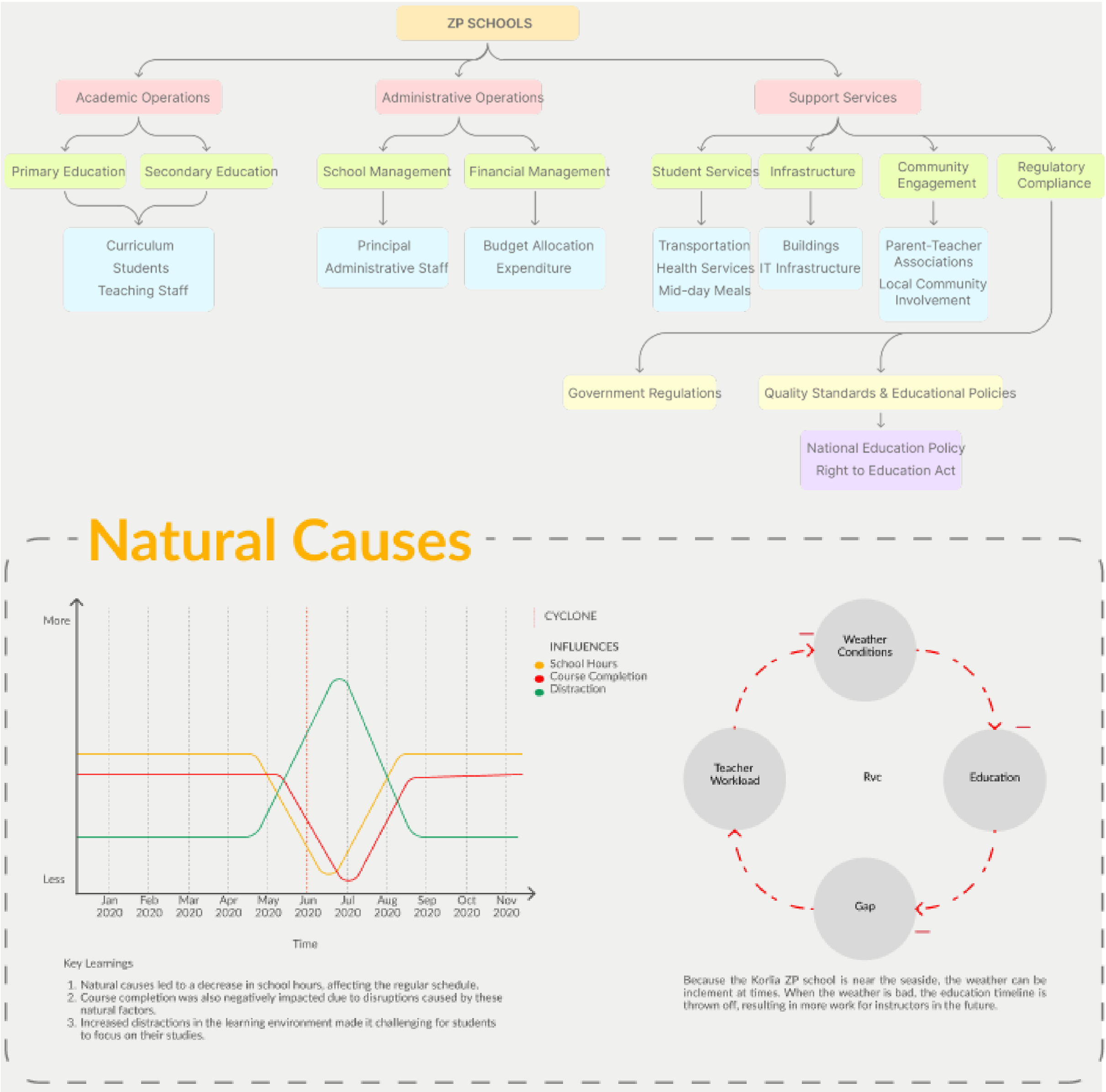
Design interventions that can reinforce learning, motivation, resource generation, and community participation over time.

05

# Ideation & exploration

The exploration focused on understanding the school through multiple system-thinking lenses before selecting interventions. Stakeholder mapping established the relationships surrounding students and teachers, while system hierarchy and element clustering exposed positive and negative factors. Causal loops, behaviour-over-time graphs, and iceberg models were then used to identify deeper relationships and hidden structures. Two leverage points emerged strongly: technology unavailability caused by environmental and rodent-related damage, and limited funds that led teachers to spend their own money on school needs.

- Mapped stakeholders across primary, secondary, and tertiary levels.
- Clustered positive and negative elements within the school system.
- Explored causal relationships through system-thinking models.
- Focused interventions on technology reliability and limited school funds.




Translating field notes into stakeholder system maps.

06

# Mechanisms & system diagrams

The system was represented through interconnected academic, administrative, support, regulatory, and community structures. Causal loops revealed reinforcing and balancing relationships, while behaviour-over-time graphs and iceberg models helped connect visible events to underlying patterns and structures.

01

## Stakeholder dependency map

Who relies on whom, for what, and where those dependencies break down.

02

### Operational friction atlas

A catalogue of friction points clustered by root cause rather than by where they surface.

07

## Prototyping & iteration

Interventions were developed around the identified leverage points and then validated with stakeholders. The proposals were refined around practical school conditions: protecting digital equipment from environmental damage, creating school-based vegetable cultivation, and using craft activities to support skills and potential local income generation.

- Developed a protective cover concept for vulnerable electronic equipment.
- Proposed vegetable cultivation within the school premises.
- Introduced craft activities as skill-building opportunities.
- Validated stakeholder acceptance of the proposed interventions.

08

## Final solution

The final direction consisted of two system-level interventions addressing the strongest identified leverage points. A protective mould cover was proposed for electronic equipment, while vegetable cultivation and craft activities were positioned as ways to engage students and support school resources.

Protective cover for school electronic equipment

Six-month responsibility rotation for maintenance

Student-led vegetable cultivation initiative

Craft and skills development activities

A shared picture of the system, built with the people inside it.

09

## Outcome & metrics

> Teachers identified potential cost savings from school-grown vegetables.
>
> Students gained opportunities for agricultural and practical skills.
>
> Craft activities introduced potential local income generation.
>
> The interventions connected school activities with community participation.

10

## Reflection

### What I learned

Systems thinking changed how I approached design problems. Instead of treating school challenges as independent issues, I learned to examine relationships, feedback loops, stakeholders, and underlying structures before deciding where an intervention could be most effective.

### What I'd improve next

I would develop longer-term validation of the interventions, particularly their effect on school resources, student participation, technology reliability, and community involvement. More sustained observation would help determine whether the proposed feedback loops actually strengthen the system over time.

## What this says about my approach

This project reflects my interest in designing beyond individual products and considering the systems that shape their use. I approach complex problems by mapping relationships, identifying leverage points, and looking for practical interventions that work with existing behaviours and resources.

Ethnographic Research / Industrial Modification

# Timber Market Machinery

An ergonomic, height-adjustable hybrid machine designed around carpenters' preferred working postures and repetitive timber-working tasks.

Project type

Ethnographic Case Study · Product Design

Role

Design researcher & modification designer

Timeline

2023-2024

Team

Research team project

Tools

Ethnographic research, Workflow mapping, Mechanical modification

Ethnographic research project

02

## The problem

Timber Market workers spend long hours performing repetitive woodworking tasks in awkward, fixed positions. The resulting physical discomfort is compounded by dusty workspaces and limited opportunities for rest. The design challenge was to support more comfortable postures while maintaining the functions required for timber turning and carving.

Improvised workflows revealed by ethnographic observation, not interviews alone.

03

## Research & insight

> **Awkward postures were a recurring concern** Workers repeatedly performed tasks in the same position for long periods, creating physical discomfort.
> **Standing was the preferred working position** A preference study with nine workers showed that the majority preferred standing across the observed processes.

> **Different processes require different positions** Cutting, turning, carving, measuring, polishing, and joining involve different working requirements.
> **Workers needed flexibility in their workspace** The research pointed toward adjustable working conditions rather than a fixed workstation posture.

04

## Design direction

The design direction focused on creating one adaptable workstation that could accommodate different woodworking activities while allowing users to adjust working height and position.

### Adjustable Working Height

The machine height can be adjusted according to the worker's convenience and ergonomic requirements.

### Hybrid Functionality

A single machine combines turning and carving functions through a rotating workstation configuration.

### Movable Hand Rest

A movable hand-rest bar supports the worker while performing the turning process.

### Simple Adjustment

A one-press locking mechanism and press nuts support straightforward adjustment and table replacement.

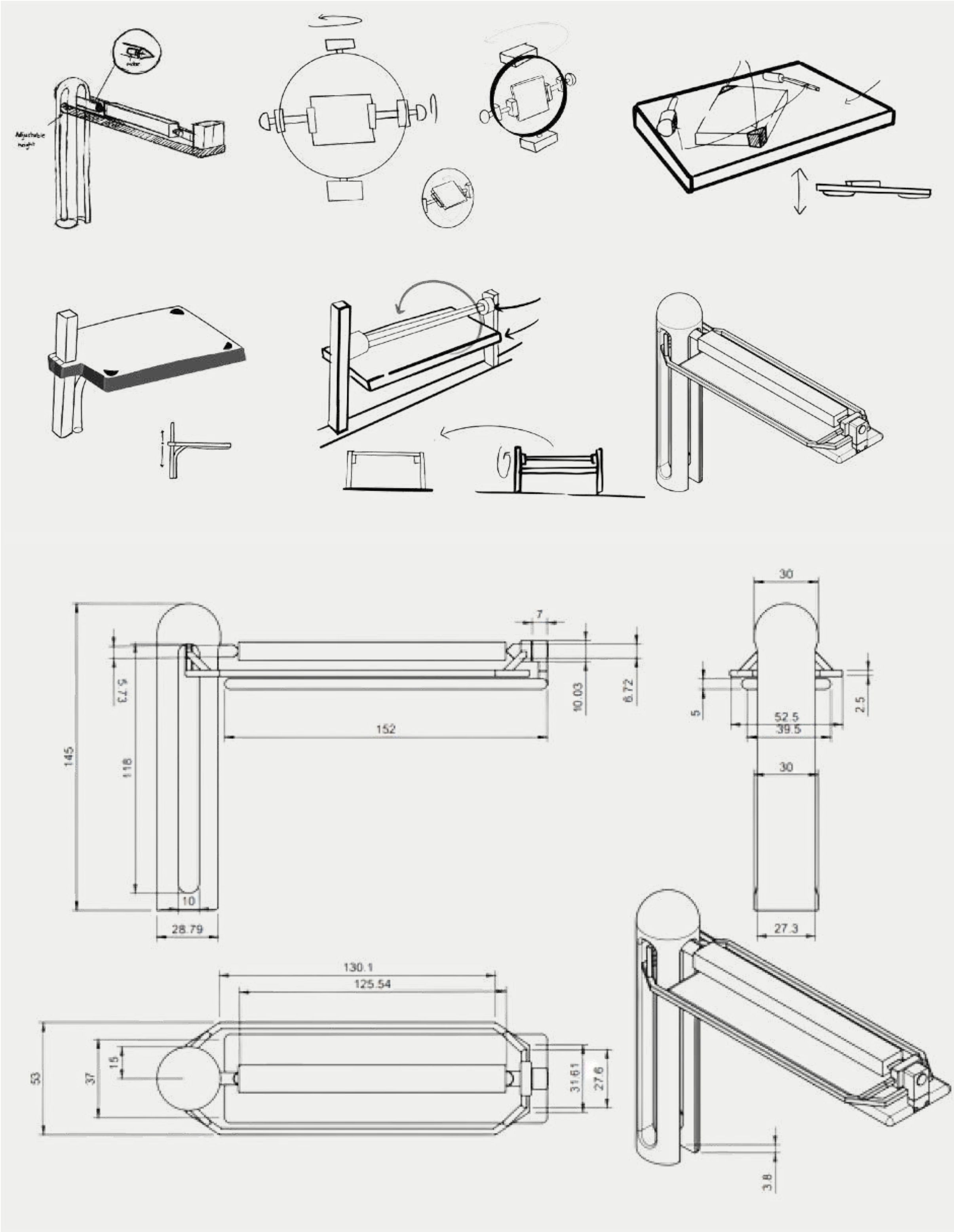
05

## Ideation & exploration

Several workstation concepts were explored around the central ergonomic problem, including an adjustable turning machine, height-adjustable worktable, hovering table, and combined turning-carving configurations. The exploration gradually focused on combining functions rather than requiring separate workstations.

The final direction developed into an adjustable turning and carving machine, incorporating a movable hand rest, rotating body, adjustable height, and interchangeable working surface.

- Adjustable turning machine with movable hand-rest bar
- Height-adjustable worktable with wood-husk collection
- Rotating workstation combining turning and carving
- Interchangeable table supported by press-nut adjustment




Modification concepts developed with, not for, the workers.

06

# Mechanisms & system diagrams

The machine brings turning and carving into one adjustable system. A central pillar supports the rotating body, while the wood holder rotates the workpiece and the workstation can be repositioned for different tasks.

01

### Motorized Wood Holder

The wood holder incorporates the motor and rotates the wooden block for the turning process.

02

### Rotating Workstation

The main body rotates around a single pillar, allowing the workstation to change configuration.

03

### Height Adjustment

The overall working height can be adjusted according to the user's convenience.

04

### Turning-to-Carving Conversion

The turning machine can rotate into a flat working desk for carving activities.

07

## Prototyping & iteration

The project developed through ergonomic and functional exploration of the workstation configuration. The design incorporated measurements based on the average Indian male height and progressively integrated adjustment, rotation, hand support, and interchangeable working surfaces.

Ergonomic dimensions were incorporated into the machine design.

Rotating configurations were explored for turning and carving.

A movable hand-rest bar was integrated into the system.

Adjustable height was incorporated for user convenience.

08

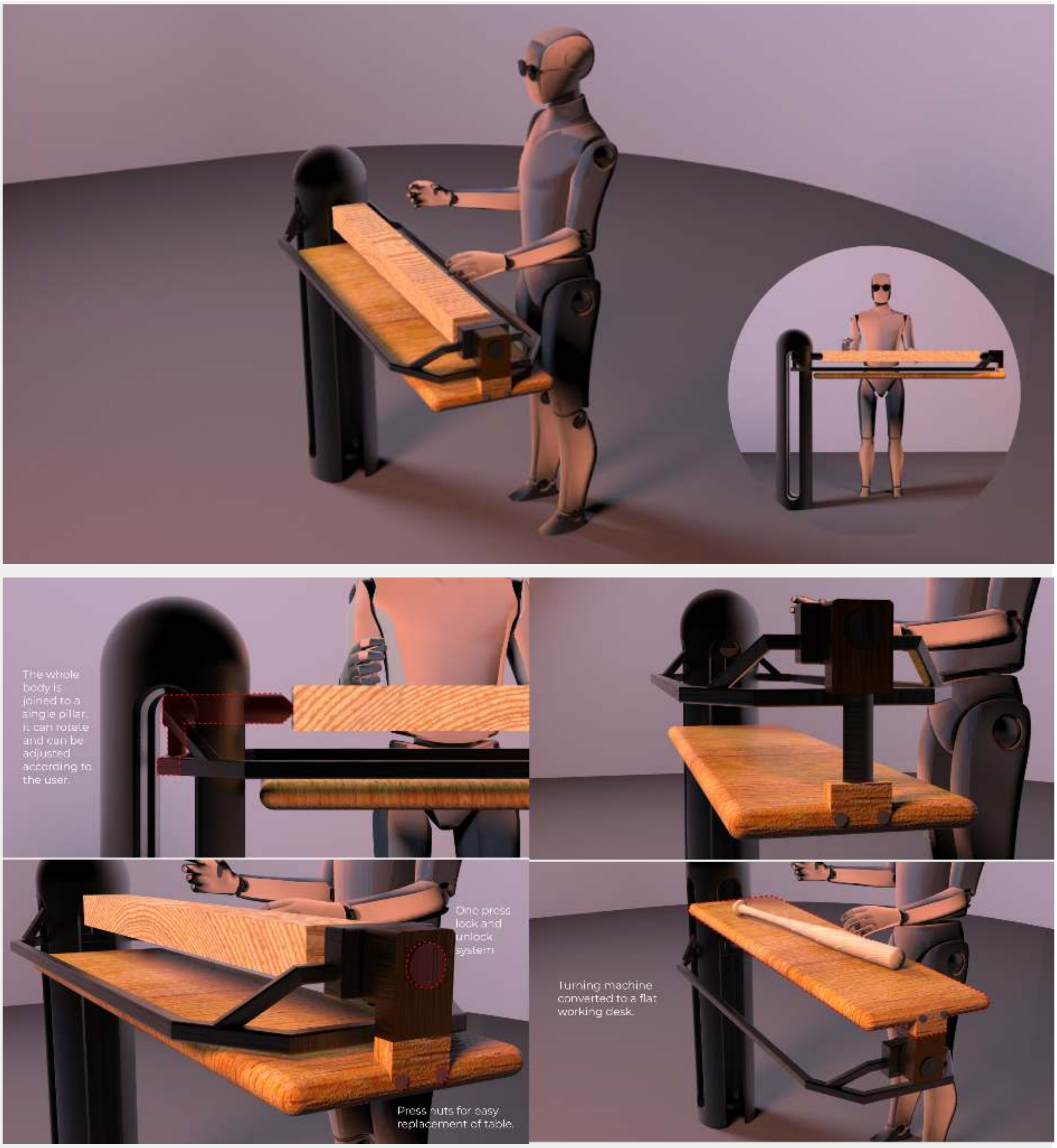
The final concept is an adjustable hybrid turning and carving machine for timber-market workers. Its rotating body, motorized wood holder, movable hand rest, and height adjustment allow the workstation to adapt between different woodworking activities.

Adjustable working height

Motorized rotating wood holder

Convertible turning-to-carving surface

Movable ergonomic hand rest




Safety that fits the workflow gets used; safety that fights it gets removed.

09

# Outcome & metrics

> Developed a hybrid turning and carving workstation.
>
> Integrated adjustable height into the machine architecture.
>
> Supported multiple woodworking activities within one system.
>
> Incorporated ergonomic considerations into the workstation design.
>
> Developed a rotating configuration for changing work modes.

# Reflection

## What I learned

The project showed how direct observation can reveal ergonomic problems that are difficult to understand from the product alone. Studying workers' actual postures and preferences helped shift the design toward an adaptable workstation rather than a fixed machine.

## What I'd improve next

I would further validate the machine with extended hands-on use, especially the adjustment mechanisms, turning-to-carving transition, and ergonomic comfort across different woodworking tasks.

## What this says about my approach

This project reflects an approach that connects ethnographic observation with product-system design. Rather than treating ergonomics as an isolated feature, the design uses adaptability and mechanical integration to respond to the user's working context.